

EXPERIMENT NO:1.(a)

AIM

To Analyze the trend of data science job postings over the last Decade

Program

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Year': list(range(2015, 2026)),
    'Job Postings': [200, 350, 500, 750, 1000, 1400, 1900, 2500, 3200, 4000,
5000]
}

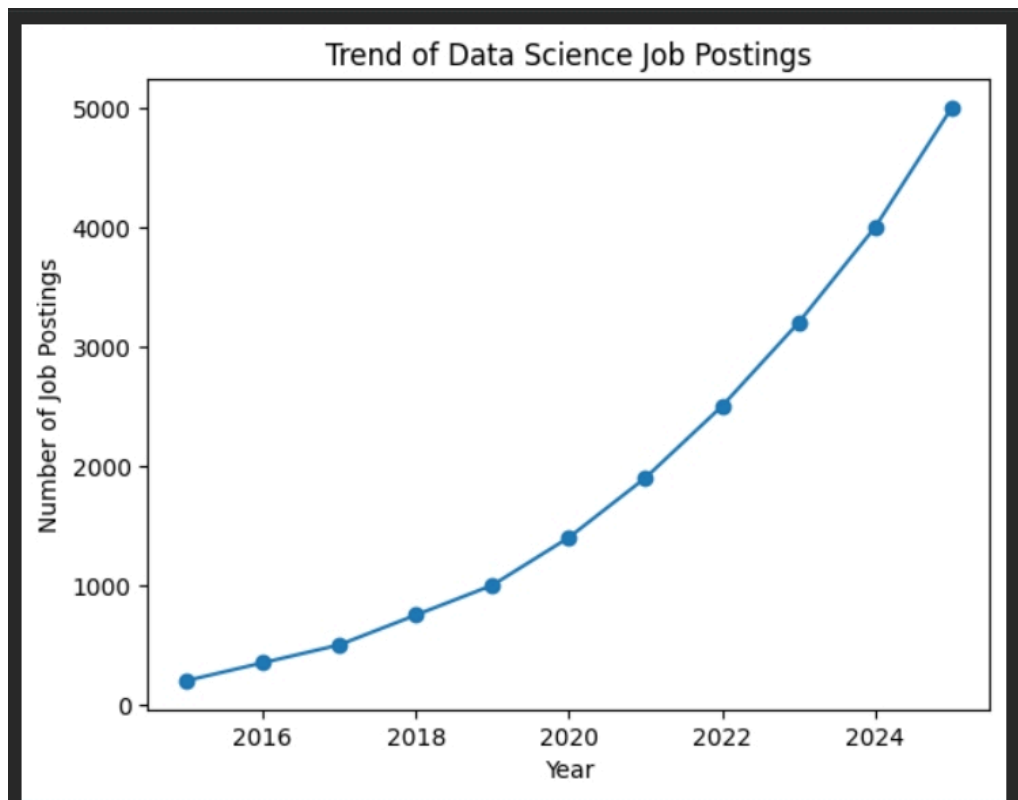
df = pd.DataFrame(data)

plt.plot(df['Year'], df['Job Postings'], marker='o')

plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')

plt.show()
```

OUTPUT



RESULT

The line graph shows an increasing trend on data science job postings from 2015 to 2025. The number of postings increased from 200 in 2015 to 5000 in 2025, a significant growth in demand for data science professionals.

EXPERIMENT NO: 1.C

AIM

To Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Program

```
import matplotlib.pyplot as plt

roles = ['Data Analyst', 'Data Engineer', 'Data Scientist',
        'ML Engineer', 'Business Analyst']

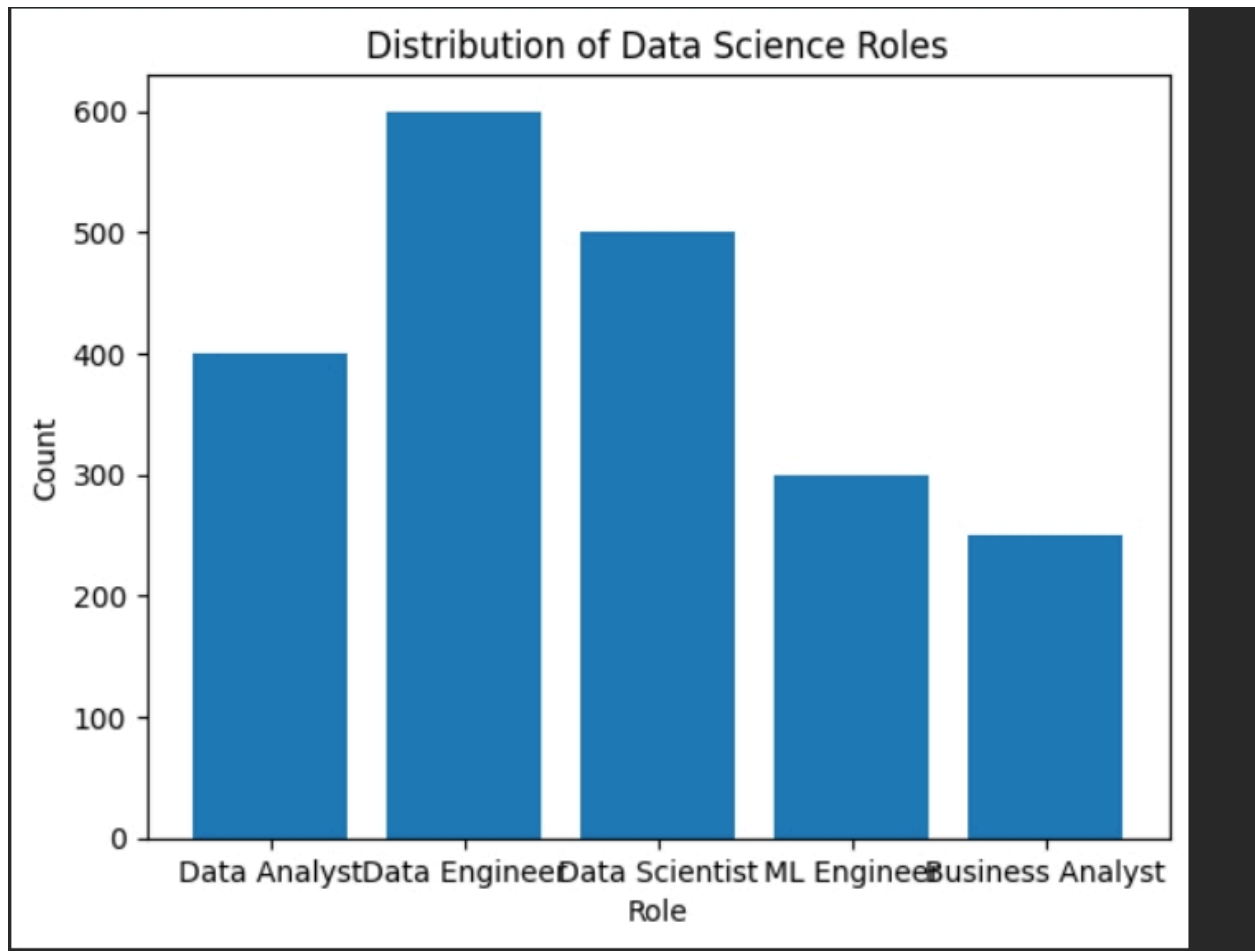
counts = [400, 600, 500, 300, 250]

plt.bar(roles, counts)

plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')

plt.show()
```

Output



RESULT

Data scientist is occurring role in the dataset (count=3) Data Analyst(3)
Data engineer(2) and ML engineer are equally dataset(count=1).

EXPERIMENT NO:1(c)

AIM

To Conduct an experiment to differentiate Structured , Un-structured and Semistructured data based on data sets given.

Program

```
import pandas as pd

# Structured data example
structured_data = pd.DataFrame({
    'ID': [101, 102, 103],
    'Name': ['David', 'Emma', 'John'],
    'Age': [28, 32, 27]
})

print("Structured Data:\n", structured_data)

# Unstructured data example
unstructured_data = "This is a sample document containing text, images
and video content."

print("\nUnstructured Data:\n", unstructured_data)

# Semi-structured data example (JSON)
semi_structured_data = {
    'ID': 101,
```

```
'Name': 'David',  
'Attributes': {  
    'Height': 170,  
    'Weight': 65  
}  
}
```

```
print("\nSemi-structured Data:\n", semi_structured_data)
```

OUTPUT

```
Structured Data:  
      ID   Name  Age  
0  101  David   28  
1  102   Emma   32  
2  103   John   27  
  
Unstructured Data:  
This is a sample document containing text, images and video content.  
  
Semi-structured Data:  
{'ID': 101, 'Name': 'David', 'Attributes': {'Height': 170, 'Weight': 65}}
```

RESULT

Structured data-(Tabular, fixed Schema) is easiest to query via sql: Semistructured JSON (xml) is self-describing but scheme flexible: Unstructured data (free text) carries no explicit schema.

EXPERIMENT NO:1.(d)

AIM

ToConduct an experiment to encrypt and decrypt given sensitive data.
Description: Use the cryptography library to encrypt and decrypt a piece of data.

Program

```
from cryptography.fernet import Fernet
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Computer Science Engineering"
cipher_text = cipher_suite.encrypt(plain_text)
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

OUTPUT

```
Original Data: b'Computer Science Engineering'
Encrypted Data: b'gAAAAABqe-zrABdJ-xAzraVuG1BR3Mzx22rU_vIUbV9-WNjvHqavQoZEUzcZH9cEvYzoCdEVakd3ZswBtJ-iB
Decrypted Data: b'Computer Science Engineering'
```

RESULT

The fernet symmetric -Relf algorithm successfully encrypted back original data text , verifying both, any tampering with the cipherset would cause decryption to fail